

Class 19 Divisibility

Goals

1. Use the division algorithm repeatedly to find $\gcd(a,b)$
2. Construct proofs involving divisibility of integers.

Recall the Division Algorithm

Definition: If a and n are integers with $n \neq 0$, we say n divides a if

other ways to say the same thing:

Notation

Examples

Definition: If a and b are nonzero integers, the greatest common divisor of a and b is

Notation

Example

Definition: Integers a and b are relatively prime (or coprime) iff

Proof Problems

#8. The divides relation is linear.

If $n \neq 0$ and $n|x$ and $n|y$, then $n|ax+by$ for any integers a & b .

Proof:

Euclidean Algorithm to find gcd (a, b)

Find the gcd (704, 407)

Reversing this process, it is always possible to find integers x and y satisfying the equation $\gcd(a,b) = ax + by$.

Example

Write 11 as a linear combination of 704 and 407.